SOLVING SCHRÖDINGER EQUATION FOR HYDROGEN ATOM USING SCILAB

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INTRODUCTION TO SCILAB :---

We cannot solve everything by ourselves on paper, some problems are even unsolvable (or very difficult to solve) for us.
So we use some computational tools to solve these kind of problems, one of them is scilab.
We can not only solve problems in scilab but can also visualize equations by plotting graphs in 2D and 3D.

BASIC MATHEMATICAL OPERATIONS :---

- To get the value of a constant, type %(constant symbol) in console.
  Example:--- %e = 2.7182818
- To write a matrix, type matrix name=[array of elemants]
  Example :--- >(1) y=[1,2,3,4,5,6,7,8,9,10] or y=[1:10]
  (2) y=[1:2:10]
  (3) a=[1 1;2 2]
- For zero matrix of order n, type A=zeros(n,n)
- For identity matrix of order n, type A=eye(n,n)
- To get inverse of a matrix, type invA.

SCHRODINGER EQUATION FOR HYDROGEN ATOM :---

$$\frac{d^2U}{dr^2} = A(r)U(r), \quad A(r) = \frac{2m}{\hbar^2}[V(r) - E] \quad \text{where } V(r) = -\frac{e^2}{r}$$

$$\frac{d^2U}{dr^2} = \frac{2m}{\hbar^2}[V(r) - E]\,U(r),$$

$e = 3.795(\text{eVÅ})^{1/2}$, $\hbar c = 1973$ (eVÅ) and $m = 0.511\times10^6$ eV/c^2

Image reference:-- https://www.google.com/url?https://www.google.com/url?s wave schrodinger equation for hydrogen atom - Google Search

GRAPH PLOTTING IN SCILAB :----

```
x=[1 2 3 4 5 6 7 8 9]
Y=[1 2 3 4 5 6 7 8 9]
plot(x,Y)
```

OBTAINING ENERGY EIGEN VALUES :----

*schrodinger eqaution in scilab 1.sce

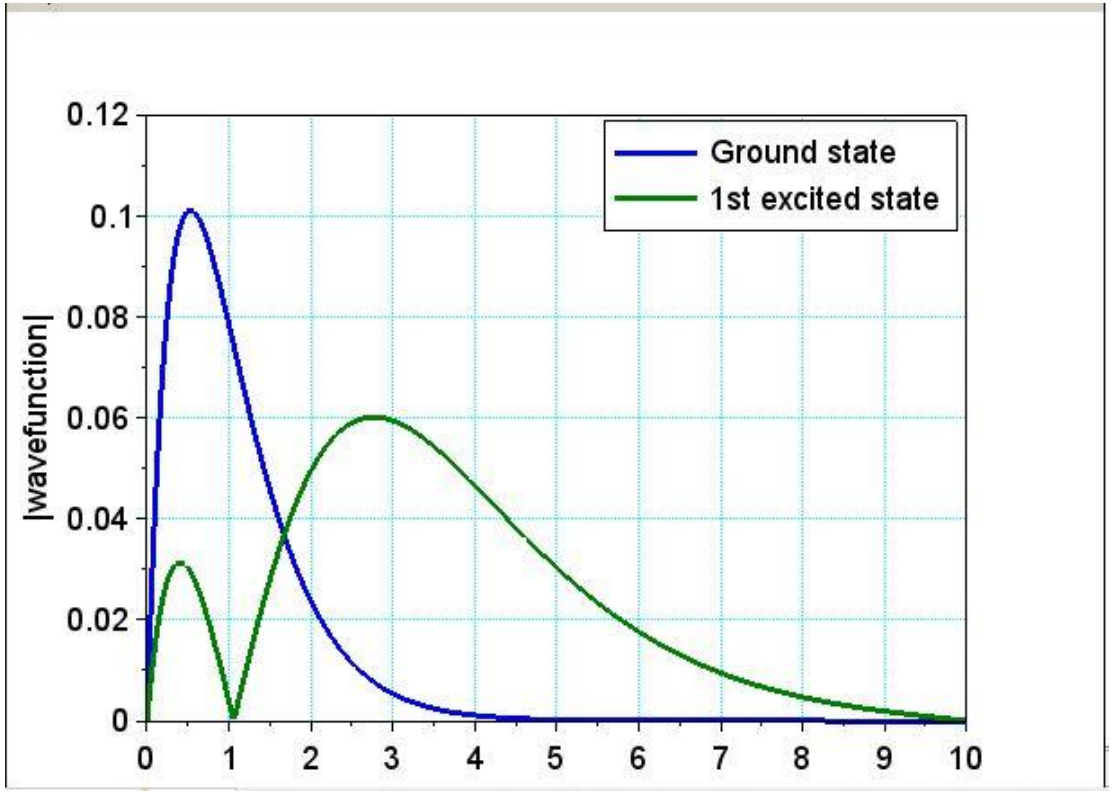
```
//solve shrodinger equation for coulomb potential
h=1973;m=0.511e6;e=3.795;//define constants
rmin=1e-10;rmax=10;n=1000;//set the range of r
r=linspace(rmin,rmax,n);//create a uniformly spaced grid of r
d=r(2)-r(1);//calculate spacing between grid pointsof r

//create potential energy matrix
V=zeros(n,n);
for i=1:n
    V(i,i)=(-(e^2)/r(i));
end
//create kinetic energy matrix
K=eye(n,n)*(-2);
for i=1:(n-1)
    K(i,i+1)=1;
    K(i+1,i)=1
end
// evaluate hamiltonian matrix H
H=(-(h^2)/(2*m*d^2))*K+V;
//calculate eigen values amd eigen vectors of H
[U,EV]=spec(H);
//extract the eigen values (at diagonal)
E=diag(EV);
//display energy eigen values
disp("ground state energy:"+string(E(2))+"eV","1st excited state energy:"+string(E(3))+"eV");
```

```
--> exec('G:\schrodinger eqaution in scilab 1.sce', -1)

  "ground state energy:-13.612722eV"

  "1st excited state energy:-3.4026106eV"
```

- **PLOTTING WAVEFUNCTION FOR HYDROGEN :---**

*schrodinger eqaution in scilab 1.sce

```
//solve shrodinger equation for coulomb potential
h=1973;m=0.511e6;e=3.795;//define constants
rmin=1e-10;rmax=10;n=1000;//set the range of r
r=linspace(rmin,rmax,n);//create a uniformly spaced grid of r
d=r(2)-r(1);//calculate spacing between grid pointsof r

//create potential energy matrix
V=zeros(n,n);
for i=1:n
    V(i,i)=(-(e^2)/r(i));
end
//create kinetic energy matrix
K=eye(n,n)*(-2);
for i=1:(n-1)
    K(i,i+1)=1;
    K(i+1,i)=1
end
// evaluate hamiltonian matrix H
H=(-(h^2)/(2*m*d^2))*K+V;
//calculate eigen values amd eigen vectors of H
[U,EV]=spec(H);
//extract the eigen values (at diagonal)
E=diag(EV);
//display energy eigen values
disp("ground state energy:"+string(E(2))+"eV","1st excited state energy:"+string(E(3))+"eV");
//plot wavefunctions
plot(r',[abs(U(:,2)),abs(U(:,3))],'linewidth',3);
xset('font size',4)
legend('Ground state','1st excited state');
xlabel('r','fontsize',4);
ylabel('|wavefunction|','fontsize',4);
xgrid(4)
```



CONCLUSION :---

- We cannot solve everything by taking a pen and a piece of paper, so we need some tools (computational tools) to solve these problems easily.
- Scilab (and other computational tools) saves so much time and hustle by solving difficult problems.
- We can visualize equations with the help of scilab.
- Solving Schrödinger equation (or any other equations) becomes easy and fun in scilab.

THANK YOU